

Low Acrylonitrile NBR

POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Abstract

The patent landscape for Low Acrylonitrile NBR (nitrile butadiene rubber) synthesis reveals a focus on emulsion polymerization processes. The major synthesis approach involves the use of various monomers, including 2-chloro-1,3-butadiene, sulfur, and optional monomers such as styrene, acrylonitrile, and methacrylonitrile. Recurring chemistry includes the use of rosin acids, metal salts of aromatic sulfonic acid-formalin condensates, and other emulsifiers. Trends in the patent landscape indicate a preference for temperatures between 30 to 55°C and conversions ranging from 70 to 90%. The resulting polymers exhibit improved properties, such as reduced heat build-up and enhanced abrasion resistance. However, some parameters, like acrylonitrile content and water amount, are not disclosed in the provided patent.

Methodology

EP:3715409:B1 - Sulfur-modified chloroprene rubber composition, vulcanized product, molded article using said vulcanized product and method for producing sulfur-modified chloroprene rubber composition

- Polymerization process: emulsion polymerization
- Acrylonitrile content: Not disclosed
- Monomer ratio: 2-chloro-1,3-butadiene and sulfur, optionally with other monomers such as styrene, acrylonitrile, methacrylonitrile, isoprene, butadiene, methacrylic acid and esters thereof
- Water amount: Not disclosed
- Emulsifier: rosin acids, metal salts of aromatic sulfonic acid-formalin condensates, sodium dodecylbenzenesulfonate, potassium dodecylbenzenesulfonate, sodium alkyl diphenyl ether sulfonate, potassium alkyl diphenyl ether sulfonate, sodium polyoxyethylene alkyl ether sulfonate, sodium polyoxypropylene alkyl ether sulfonate, potassium polyoxyethylene alkyl ether sulfonate, and potassium polyoxypropylene alkyl ether sulfonate
- Initiator: potassium persulfate, benzoyl peroxide, ammonium persulfate, hydrogen peroxide
- Chain transfer agent: Not disclosed
- Temperature: 0 to 55°C, preferably 30 to 55°C
- Reaction time: Not disclosed
- Conversion: 60 to 95%, preferably 70 to 90%

- Coagulation: Not disclosed
- Polymer properties: Mooney viscosity of 20 to 80, improved permanent compression set and abrasion resistance, reduced heat build-up

Cross-Patent Comparison & Synthesis Trends

Emulsifier Selection and Loading

The use of rosin acids, metal salts of aromatic sulfonic acid-formalin condensates, and other emulsifiers is a common trend in the patent landscape.

Initiator and Chain Transfer Agent Strategies

The patent utilizes potassium persulfate, benzoyl peroxide, ammonium persulfate, and hydrogen peroxide as initiators, while the chain transfer agent is not disclosed.

Monomer Ratio and Reaction Control

The monomer ratio involves 2-chloro-1,3-butadiene and sulfur, with optional monomers, and the reaction control is achieved through temperature and conversion management.

Coagulation and Post-Treatment Conditions

The coagulation conditions are not disclosed in the provided patent.

References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year	Google Patents URL
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EP:3715409:B1	Sulfur-modified chloroprene rubber composition, vulcanized product, molded article using said vulcanized product and method for producing sulfur-modified chloroprene rubber composition	Denka Co Ltd	EP	2021	https://patents.google.com
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